

NATURAL LANGUAGE PROCESSING INTERNSHIP

Task: 1

Author:
Malak Abdelakher

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1 Dataset Refinement and Quality Improvement

An initial inspection of the dataset revealed several quality issues that could negatively affect model training. Although the dataset contained 3,000 query-image pairs, analysis showed that the *image descriptions* column contained only 12 unique descriptions, meaning that many unrelated queries were paired with identical image captions. Such limited diversity prevented the model from learning meaningful relationships between textual queries and their associated visual descriptions.

To address this limitation, a multi-stage dataset refinement pipeline was developed using a locally deployed Large Language Model (LLM) through Ollama.

1.1 Generation of Image Descriptions

After removing duplicate queries, the dataset size was reduced to 2,009 unique samples. For each query, the LLM was prompted to generate a short, generic, and non-graphic image description that was semantically consistent with the corresponding query while avoiding explicit or harmful content. The generated descriptions served as textual representations of images, increasing the diversity of the multimodal information available for training.

Certain sensitive categories, such as *Child Sexual Exploitation*, were excluded from automatic generation and instead assigned a fixed placeholder description to ensure ethical handling of the data.

1.2 Correction of Refused Generations

During the generation process, the LLM refused to answer many prompts because they involved harmful or sensitive topics. Approximately 626 generated descriptions contained refusal responses (e.g., "I cannot assist with that request"), which were unsuitable for training.

To overcome this issue, a post-processing step was introduced. Refusal responses were automatically detected using predefined refusal patterns and replaced with manually designed, category-specific template descriptions. These templates remained neutral and non-graphic while preserving the semantic context of each toxicity category.

1.3 Label Verification

Since the original dataset labels were manually assigned, an additional verification stage was performed to identify potential annotation errors. Each query was reclassified using the LLM based on predefined category definitions. The predicted category was then compared with the original dataset label.

Rather than automatically modifying the dataset, all disagreements were exported to a separate review file for manual inspection. Out of the 2,009 samples, the LLM agreed with the existing labels for 536 instances, while 1,473 samples were flagged for manual review. This conservative approach ensured that the final dataset maintained human oversight and avoided introducing incorrect automatic label corrections.

1.4 Class Balancing

The refined dataset exhibited severe class imbalance, with categories such as *Suicide & Self-Harm*, *Elections*, and *Sex-Related Crimes* containing only a few samples.

To improve class representation, data augmentation techniques were applied to these minority classes. New query variations were generated using synonym replacement and lexical transformations while preserving the original semantic meaning. Categories involving child sexual exploitation were intentionally excluded from augmentation due to ethical considerations.

Following augmentation, the minority classes were expanded to approximately 20 samples each, resulting in a more balanced training dataset while maintaining the integrity of the original annotations.

1.4.1 Lessons Learned and Dataset Transition

Despite the improvements introduced through data cleaning, description generation, label verification, and class balancing, several practical limitations remained. The most significant challenge arose from the use of Large Language Models (LLMs) to generate image descriptions for sensitive categories. Due to built-in safety mechanisms, the LLM frequently refused to generate descriptions for queries related to highly sensitive content, particularly *Child Sexual Exploitation* and *Sex-Related Crimes*. Although template-based replacements were

introduced to mitigate this issue, these descriptions lacked the diversity and contextual richness required for effective multimodal learning.

Furthermore, the extensive manual review required to verify generated descriptions and validate category labels significantly increased the data preparation effort. The combination of limited data diversity, severe class imbalance, and restrictions imposed by the LLM made it difficult to construct a reliable and representative dataset for training.

Consequently, it was decided to discontinue this dataset and adopt an alternative benchmark dataset that provided higher-quality annotations, better class representation, and more diverse multimodal content. This transition enabled subsequent experiments to focus on model development and evaluation using a more robust and reliable dataset while avoiding the limitations encountered during the initial data preparation process.

2 Multi-Label Toxic Comment Classification: RNN and LSTM Results

2.1 Dataset and Preprocessing

The Jigsaw Toxic Comment Classification dataset was used, consisting of 159,571 labeled comments across six binary, non-mutually-exclusive labels: `toxic`, `severe_toxic`, `obscene`, `threat`, `insult`, and `identity_hate`. The dataset contained no missing values and no duplicate rows.

2.2 Data Splitting

A multilabel-stratified split (`MultilabelStratifiedShuffleSplit`) was used to preserve label frequency across splits, applied *before* augmentation so validation and test sets contain only original, unaugmented text.

Split	Rows	% of Total
Train	111,699	70.00%
Validation	23,936	15.00%
Test	23,936	15.00%

Table 1: Stratified train/validation/test split.

2.3 Data Augmentation

Rare classes (`severe_toxic`, `threat`, `identity_hate`) were augmented using WordNet-based synonym substitution alternated with random word swap (via `nlpaug`), applied only to the training set.

Label	Before Augmentation	After Augmentation
<code>toxic</code>	10,706	31,314
<code>severe_toxic</code>	1,116	11,446
<code>obscene</code>	5,914	22,317
<code>threat</code>	335	8,951
<code>insult</code>	5,514	21,957
<code>identity_hate</code>	983	10,824

Table 2: Label counts before and after augmentation.

2.4 Tokenization and Vocabulary

A custom regex tokenizer (lowercasing, URL removal, non-alphanumeric stripping) was used. Vocabulary was built from the augmented training set only, capped at 30,000 tokens plus `<PAD>`/`<UNK>`, yielding a final vocabulary size of 30,002. Sequences were truncated/padded to `MAX_LEN = 150` tokens (95th percentile comment length: 231 tokens; median: 33 tokens).

2.5 Model Architectures

For both RNN and LSTM families, a single flexible classifier class (PyTorch `nn.RNN` / `nn.LSTM`) was implemented with `num_layers`, `bidirectional`, and `dropout` as configurable parameters, enabling seven architecture variants per family without duplicating model code.

Architecture	Layers	Bidirectional	Dropout
Simple	1	No	0.0
Stacked	2	No	0.0
Deep	3	No	0.0
Bidirectional	1	Yes	0.0
Stacked Bidirectional	2	Yes	0.0
With Dropout	1	No	0.3
Best-config (RNN: Stacked+Dropout; LSTM: Bidirectional+Dropout)	1-2	Varies	0.3

Table 3: Seven architecture configurations trained per family (RNN and LSTM).

2.6 Training Configuration

Both families were trained with the following shared setup: embedding dimension 128, hidden dimension 128, `BCEWithLogitsLoss` with per-class `pos_weight` (clipped to $[1, 10]$) to counteract class imbalance, Adam optimizer ($\text{lr} = 1 \times 10^{-3}$, weight decay 1×10^{-5}), gradient clipping at max norm 5.0, batch size 64, 8 epochs per architecture, and a fixed decision threshold of 0.5 on sigmoid outputs. The checkpoint with the highest validation F1 Macro was restored after training for each architecture.

2.7 RNN Results

Architecture	F1 Macro	F1 Micro	F1 Weighted
Simple_RNN	0.5934	0.7221	0.7284
Stacked_RNN	0.5872	0.7109	0.7175
Deep_RNN	0.5727	0.7045	0.7250
Bidirectional_RNN	0.5897	0.7187	0.7314
Stacked_BiRNN	0.5795	0.7108	0.7186
RNN_Dropout	0.5891	0.7141	0.7223
Stacked_RNN_Dropout	0.6044	0.7253	0.7304

Table 4: RNN architecture comparison on the validation set. Best architecture: Stacked_RNN_Dropout.

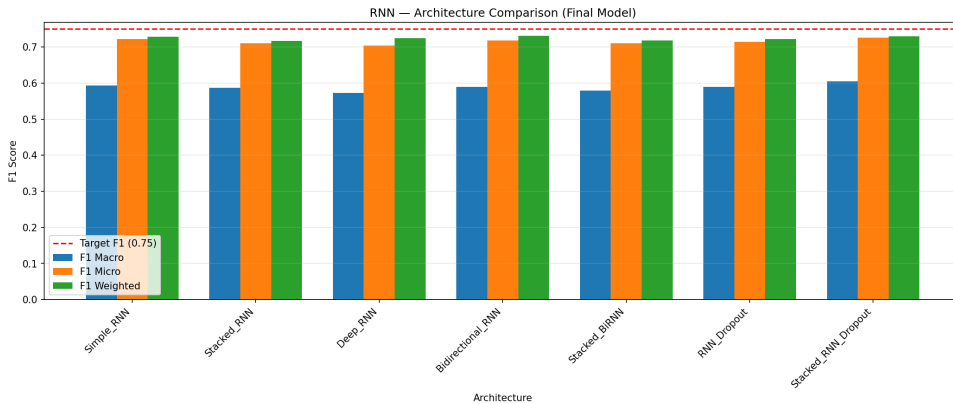


Figure 1: RNN architecture comparison — validation F1 Macro/Micro/Weighted per configuration.

RNN Task Requirement ($F1 \geq 0.75$): The best RNN model achieved an F1 Macro of 0.6044, which does not meet the required threshold of 0.75. F1 Micro (0.7253) and F1 Weighted (0.7304) were closer but also below target.

RNN Training Curves

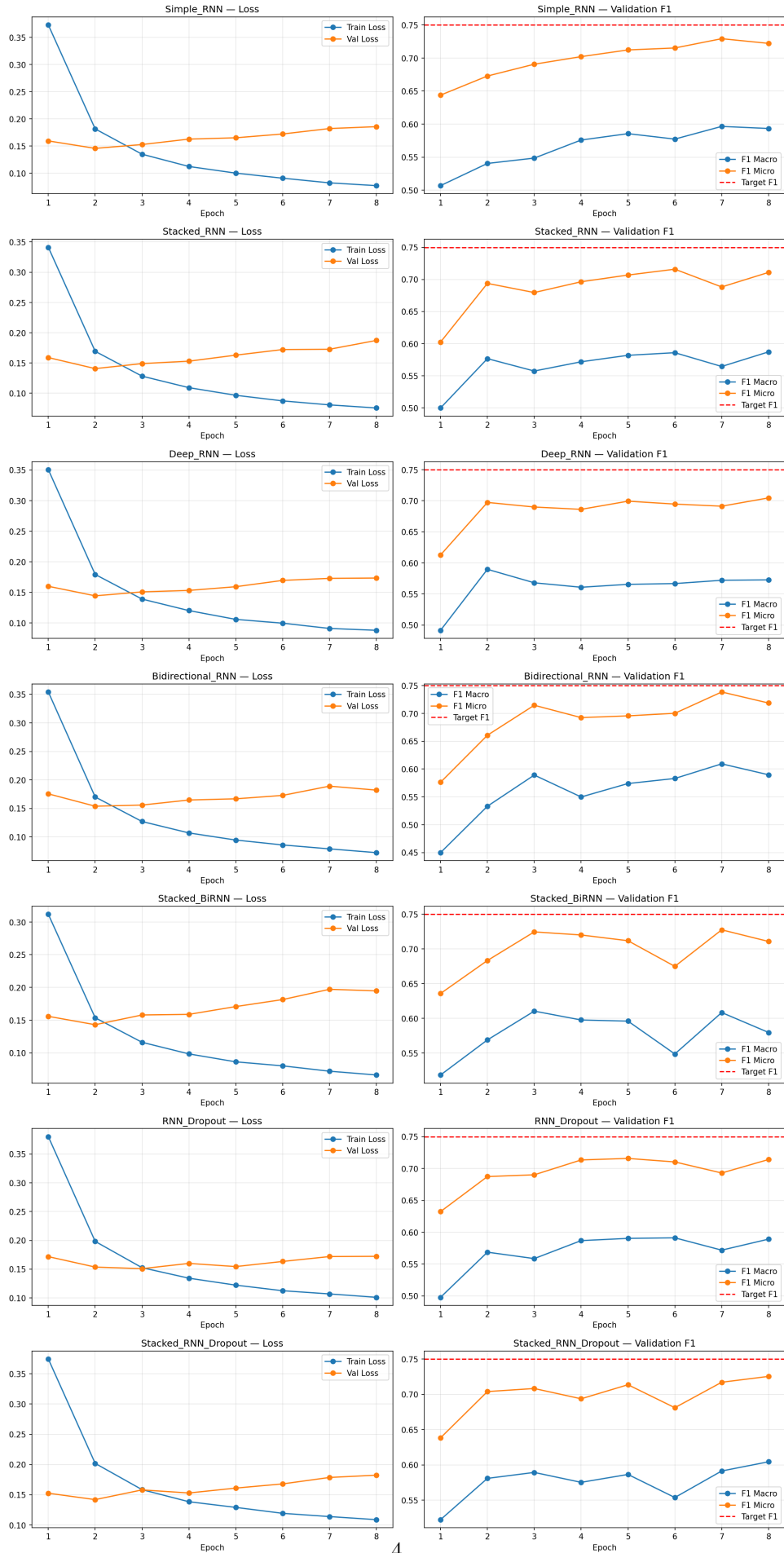


Figure 2: RNN training and validation loss, and validation F1, per epoch, across all seven architectures.

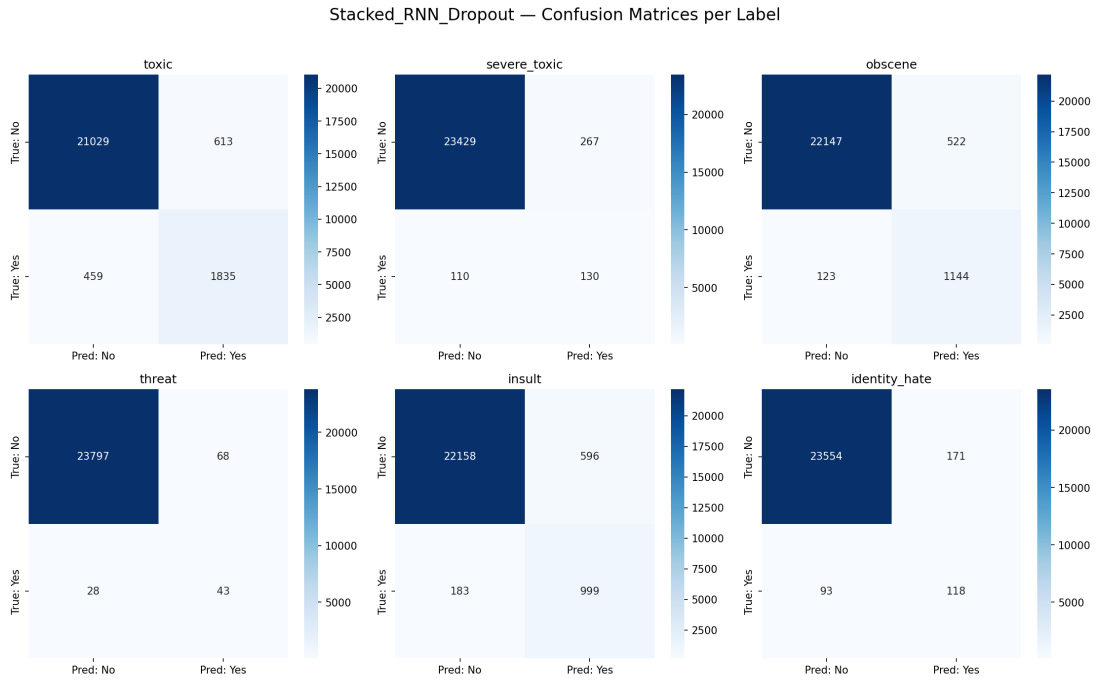


Figure 3: Per-label confusion matrices for the best RNN model (Stacked_RNN_Dropout).

2.8 LSTM Results

Architecture	F1 Macro	F1 Micro	F1 Weighted
Simple_LSTM	0.5821	0.7183	0.7274
Stacked_LSTM	0.5831	0.7114	0.7165
Deep_LSTM	0.6054	0.7358	0.7417
Bidirectional_LSTM	0.6116	0.7402	0.7439
Stacked_BiLSTM	0.5990	0.7302	0.7393
LSTM_Dropout	0.5904	0.7282	0.7415
BiLSTM_Dropout	0.6044	0.7385	0.7411

Table 5: LSTM architecture comparison on the validation set. Best architecture: Bidirectional_LSTM.

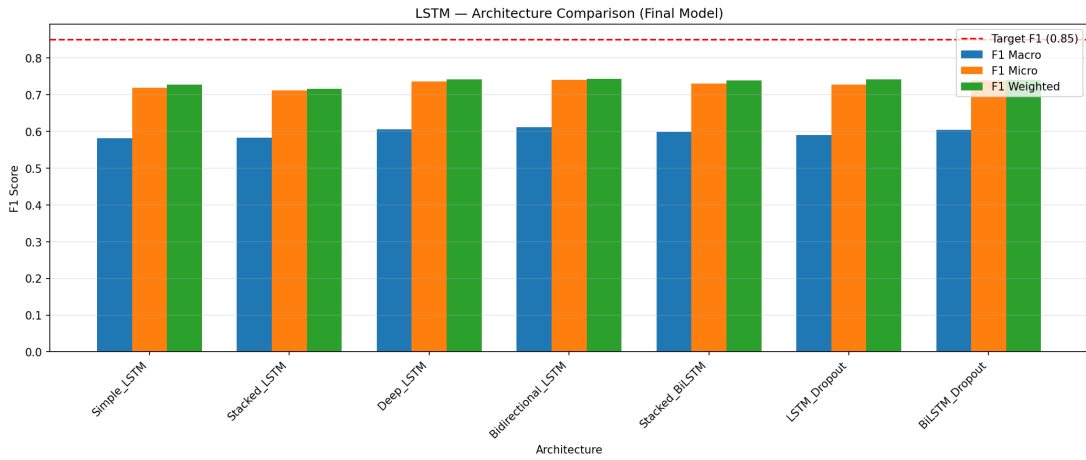


Figure 4: LSTM architecture comparison — validation F1 Macro/Micro/Weighted per configuration.

LSTM Task Requirement ($F1 \geq 0.85$): The best LSTM model achieved an F1 Macro of 0.6116, which does not meet the required threshold of 0.85. F1 Micro (0.7402) and F1 Weighted (0.7439) were notably higher but also below target.

2.9 Limitations and Recommended Next Steps

F1 Macro weighs all six labels equally, so weak recall on rare classes (`threat`, `identity_hate`, visible in the confusion matrices above) disproportionately lowers the macro average even though frequent classes (`toxic`, `obscene`, `insult`) perform well. Identified causes and recommended fixes:

- **No pretrained embeddings.** Word embeddings were randomly initialized; loading pretrained GloVe or Word2Vec vectors would likely improve rare-word semantics given the comparatively small training set.
- **Fixed 0.5 decision threshold.** Rare classes rarely produce probabilities near 0.5 even for correct positive predictions; per-class threshold tuning on the validation set was identified but not yet applied.
- **Limited training epochs.** Validation loss begins rising (mild overfitting) around epoch 3–4 for most configurations, suggesting a combination of stronger regularization, a learning-rate scheduler, and early stopping would allow more effective use of additional epochs.
- **Further augmentation or focal loss.** Increasing augmentation volume specifically for `threat/identity_hate`, or replacing/augmenting the weighted BCE loss with focal loss, would more directly target the hard, rare examples driving the macro F1 gap.

LSTM Training Curves

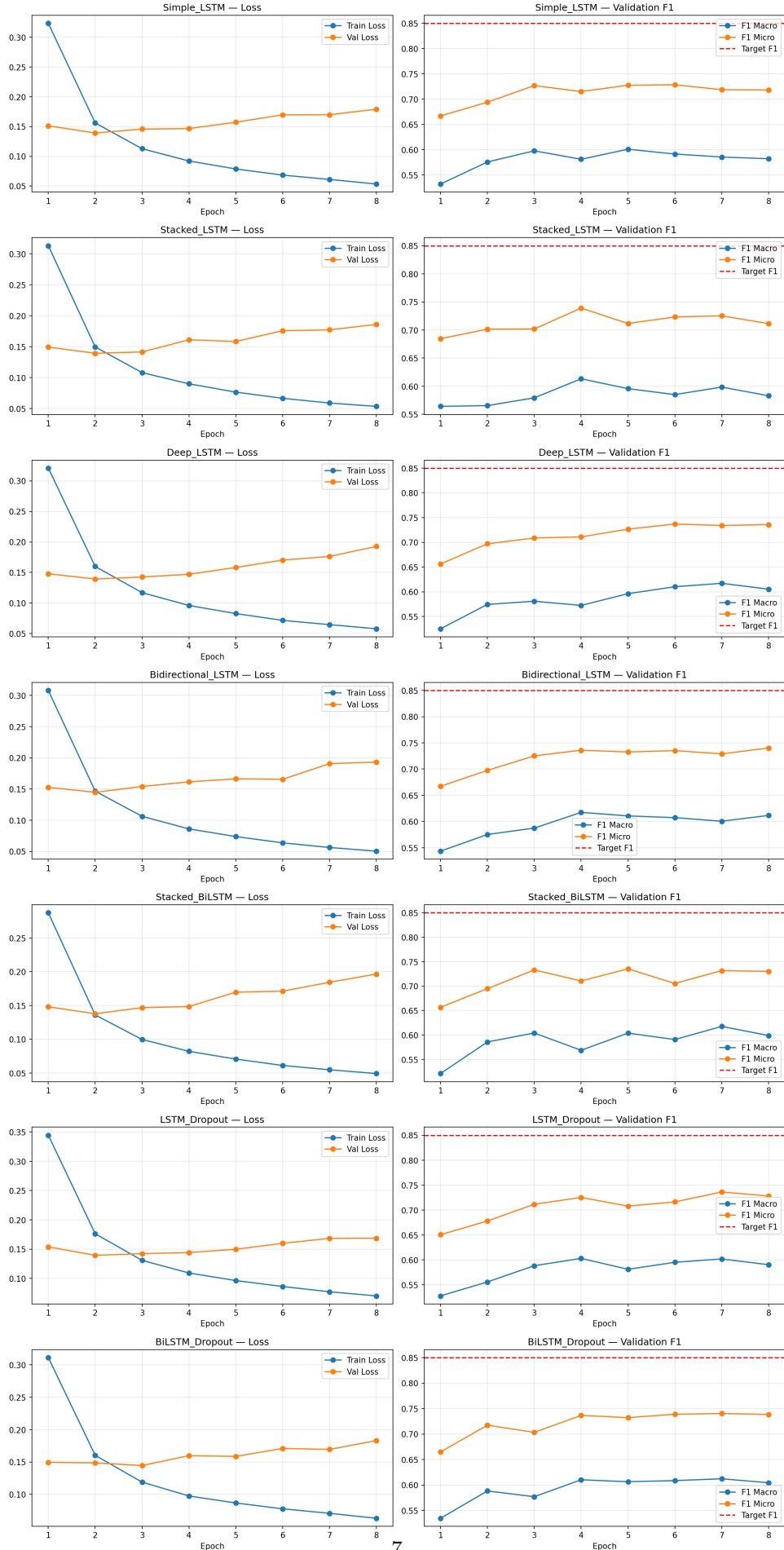


Figure 5: LSTM training and validation loss, and validation F1, per epoch, across all seven architectures.

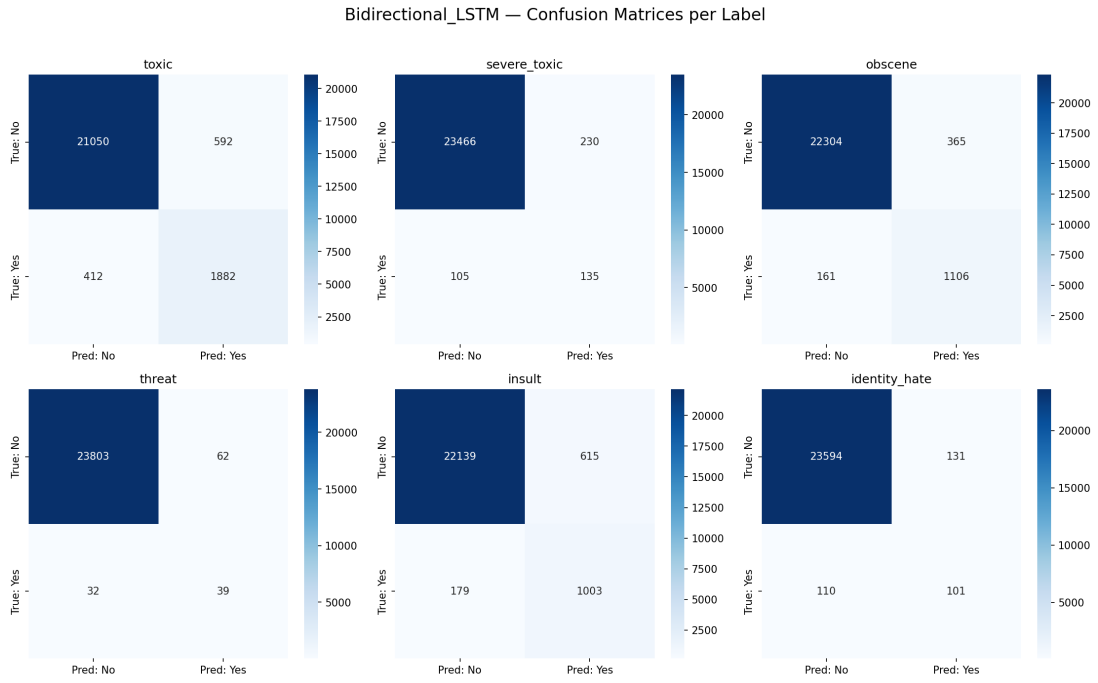


Figure 6: Per-label confusion matrices for the best LSTM model (Bidirectional_LSTM).